PHOL0001
Mammalian Physiology
Introduction



Organiser: Dr Pam Houston
p.houston@ucl.ac.uk

Deputy Organiser: Dr Jigisha Patel
Jigisha.a.patel@ucl.ac.uk

This session is concerned with the arrangements for:

PHOL0001 Mammalian Physiology

30 Credits, FHEQ level 4

taken by

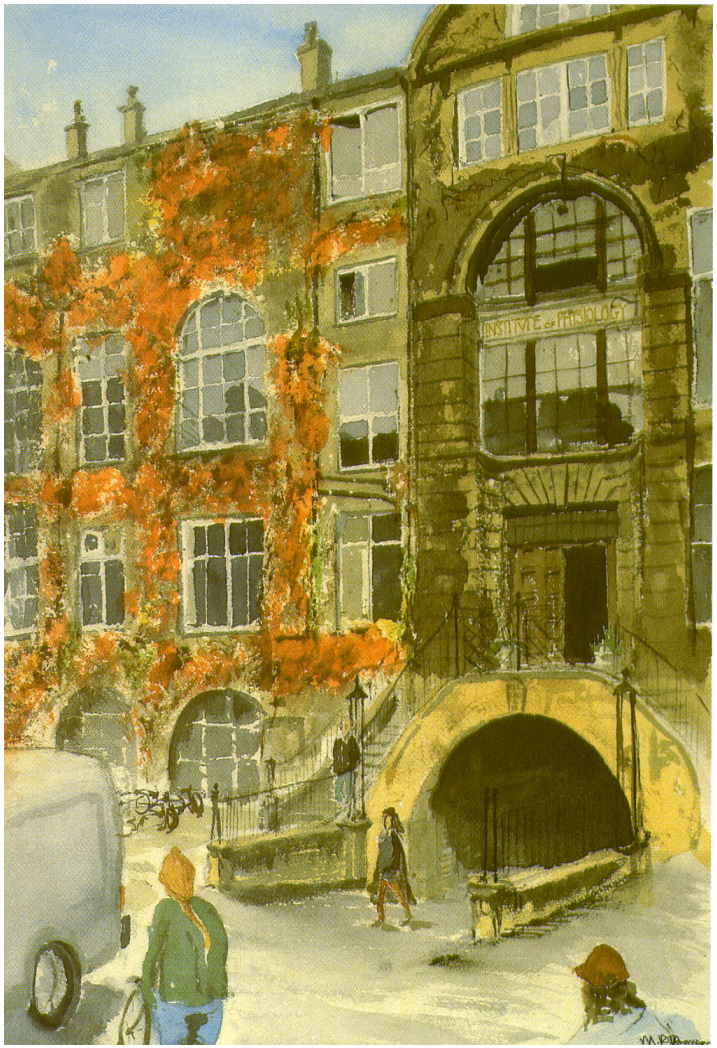
BSc Biomedical Sciences

BSc/MSci Pharmacology

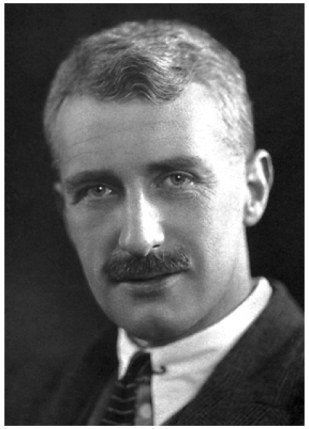
Affiliates

30 credits equates to 15 ECTS or 300 learning hours.
A full-time academic year equates 120 credits, 60 ECTS and 1200 learning hours.

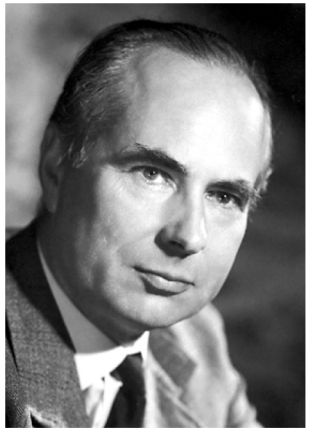
- *Physis* meaning “nature” or “origin” and *logia* “study of” ….the study of the functions of living matter.
- Is concerned with how an organism performs its varied activities: how it feeds, how it moves, how it breathes, how it adapts to changing circumstances, how it spawns new generations.
- The success of physiology in explaining how organisms work is based on the notion that they are intricate and exquisite machines whose operation is governed by the laws of Physics and Chemistry.
- To study how an organism works it is first necessary to know how it is built. A full appreciation of the physiology of an organism must therefore be based on a sound knowledge of its Anatomy.



The Department of Physiology at UCL was one of the oldest in the UK. Now part of the Neurophysiology, Pharmacology and Physiology (NPP) Research Department. Out of 29 Nobel Laureates who have worked at UCL, 10 were associated with the Physiology Dept. including AV Hill, Andrew Huxley, and Bernard Katz.



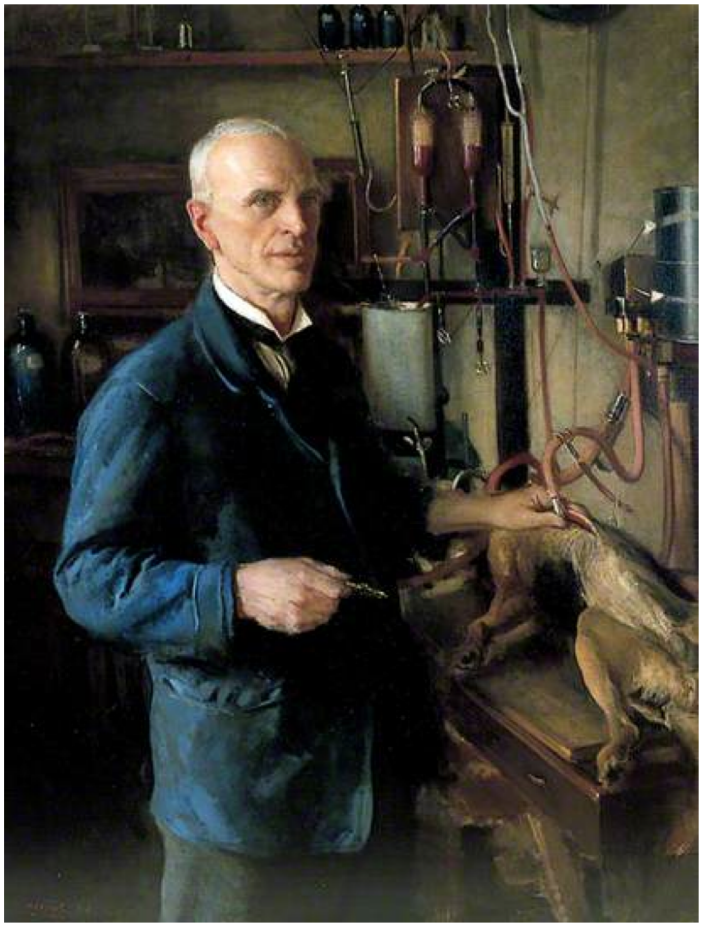
Archibald V. Hill



Andrew Huxley



Bernard Katz



E.H Starling by Walter Westley Russell
© the copyright holder photo credit UCL Art Museum

Ernest Starling 1866-1927

Major contributions to physiology were:

- Devised the Frank-Starling law of the heart
- Discovered secretin the first hormone
- The Starling equation (Starling's forces)
- The discovery of peristalsis
- Demonstrated that the convoluted tubule of the kidney reabsorbs water and electrolytes

- Award jointly with Professors May-Britt Moser and Edvard Moser of the Norwegian University of Science and Technology.
- "for their discoveries of cells that constitute a positioning system in the brain."
- Discovered the first component of this positioning system in 1971.
- These "place cells" that are located in a part of the brain called the hippocampus that builds up an inner map of the environment.

Professor John O'Keefe
Nobel Prize for Physiology or Medicine 2014



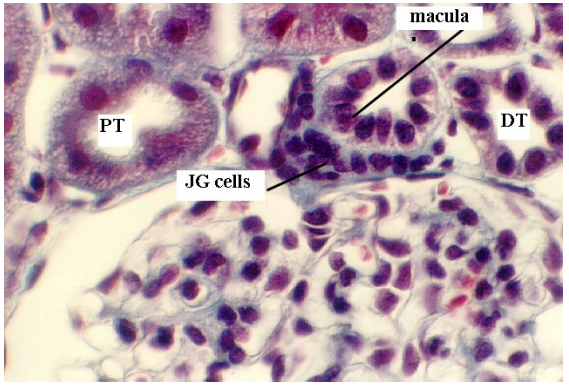


The building blocks of the body are the cells, which are grouped together to form tissues.

The principal types of tissue are:

- Epithelial
- Connective (fibrous tissues)
  - Blood
  - Bone
- Nervous
- Muscular





http://www.cytochemistry.net/microanatomy/kidney/kidney.htm

Organs are formed by the aggregation of different kinds of tissue.

Examples:

- Brain is made up of connective, epithelial, and nervous tissues
- Heart is made up of connective, epithelial and muscular tissue plus some nerve ganglia





https://commons.wikimedia.org/wiki/File:CG_Heart.gif

The organs are themselves parts of distinct physiological systems.

The Principal Organ Systems

- The musculo-skeletal system
- The nervous system
- The endocrine system
- The respiratory system
- The renal system
- The immune system
- The cardiovascular system
- The gastro-intestinal system

Topics

• Cellular physiology	Dr P Houston & Dr J Patel
• Nerve and muscle	Dr J Patel & Prof M Mayston
• The respiratory system	Dr J Patel
• Endocrinology	Dr P Houston
• The kidney and the renal system	Dr P Houston
• The immune system	Dr J Patel
• The cardiovascular system	Prof G Szabadkai
• The nervous system	Prof M Mayston
• Integrative aspects of physiology	Dr J Patel
• The gastro-intestinal system	Dr J Marks

By the end of the module you should have:

- Acquired a basic knowledge of the normal functioning of the body, it's tissues, organs and systems
- Some knowledge of the methods and principles of measurement of physiological functions from the practical classes
- Learn how to assess scientific evidence and analyse data
- Discovered how and where to find further information
- Learn to communicate clearly and logically and collaborate effectively
- Used the material provided on Moodle to complete the set assignments and exercises, including mathematical skills.

The module consists of:

- 40 Lectures
- Four Practical classes
- LT learning platform, pre-practical content
- Post-practical quizzes
- Four in-course assessments with live scores, re-attempts and feedback
- An online 3 hour end of of year exam
- Plus 12 tutorial events.

- Are held on Mondays and Wednesdays
- Rooms can change, therefore it is important to regularly check the UCL's online timetable
- Lecture slides will be posted on Moodle beforehand
- Zoom will be used to host these events, if not delivered face to face
- Lectures will be recorded and files posted on Moodle, where possible.
- Don't rely on Panopto recordings!

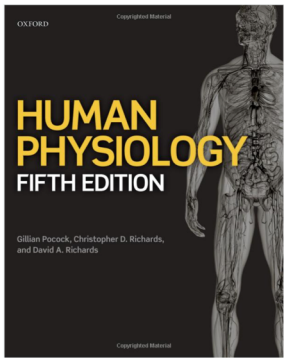
- The recorded lectures and workshops will not just be a list of facts to be learnt; they will often present a scientific argument, which you are expected to grasp
- They are **NOT** intended to be your sole source of information!
- You are also expected to read the relevant pages in the recommended textbook and do your own private study
- Lecturers do not dictate notes for you to learn
  - You therefore need to take good notes
  - Make a list of the main headings during the lecture/workshop/practical and afterwards use your reading to fill in the missing details.

The recommended textbook is:

Human Physiology (fifth edition) by G Pocock, CD Richards & D Richards, OUP (2018).

Copies can be bought from Waterstones, borrowed from a UCL library or accessed online

Editions 3 and 4 are still available and acceptable

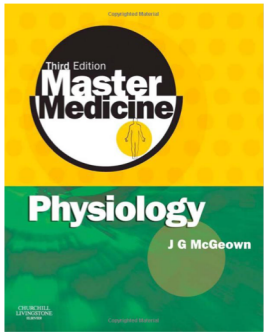


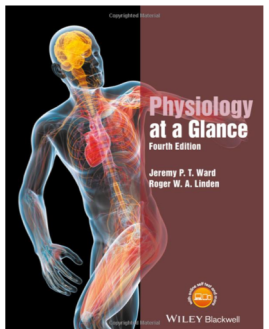


More basic texts:

Physiology by J.G. McGeown, published by Churchill-Livingstone (2001/2007)

Physiology at a glance by Ward and Linden publish by Blackwell (2017).

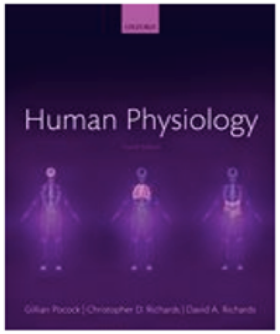




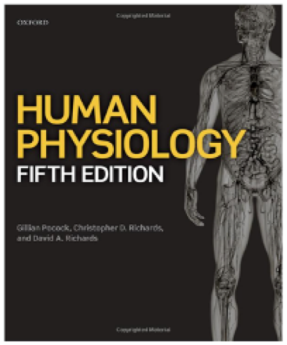
Human Physiology: Pocock, Richards and Richards



3rd Edition:
Chapter 2, pages 5-16



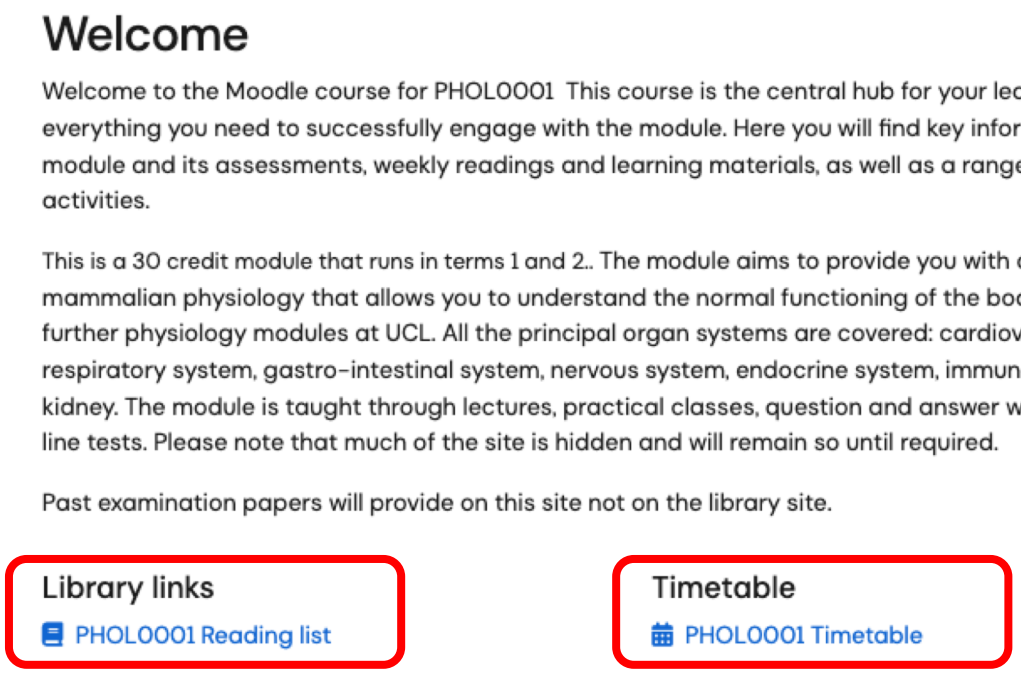
4th Edition:
Chapter 2, pages 14-25
Chapter 3, pages 27-43



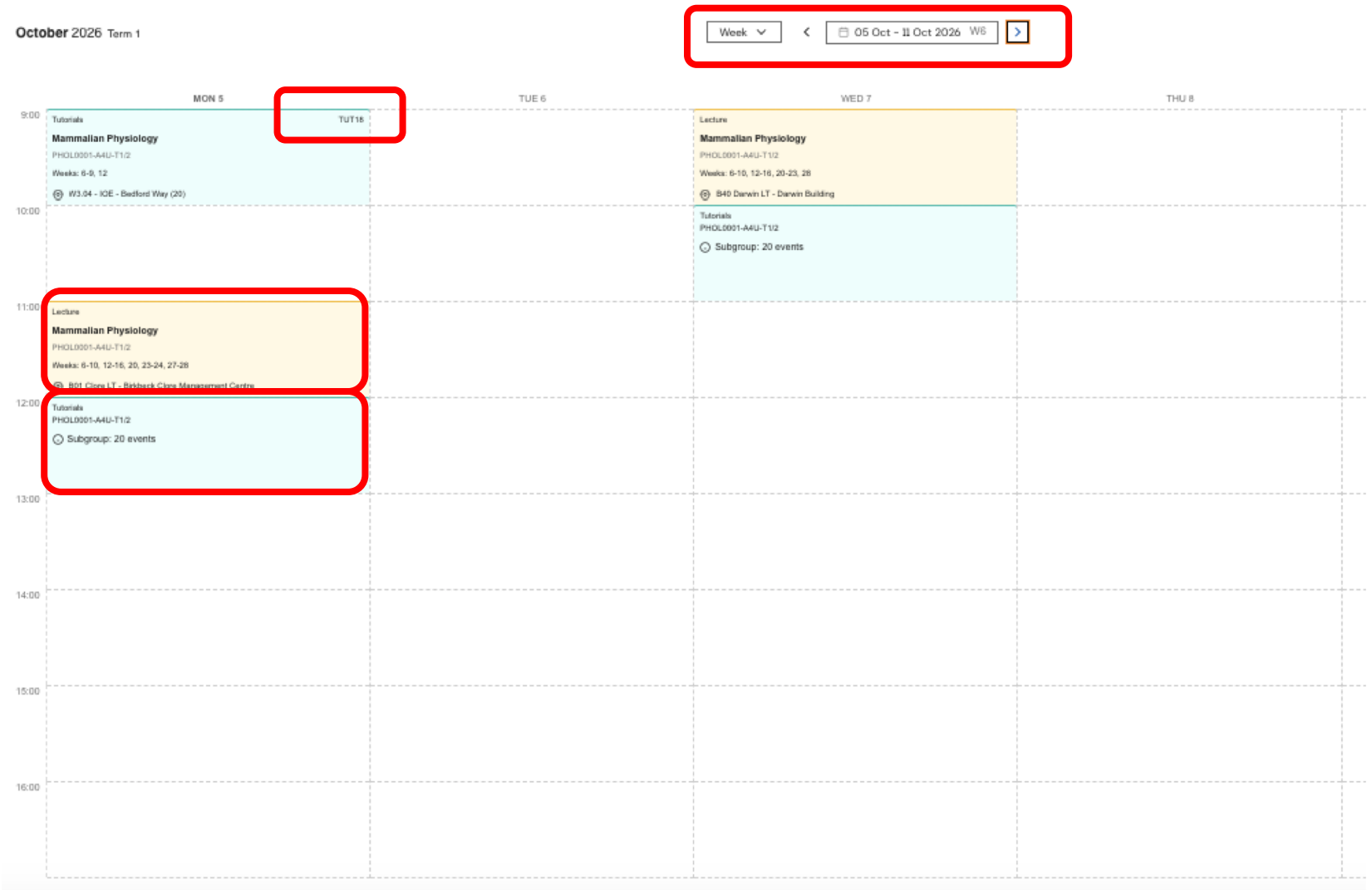
5th Edition:
Chapter 2, pages 14-25
Chapter 3, pages 27-43

Timetable

- Link is available on Moodle, or
- The UCL online timetable via your personal, degree or custom settings.



Timetable


October 2026 Term 1
Week
05 Oct – 11 Oct 2026 W6
MON 5
TUE 6
WED 7
THU 8
9:00
10:00
11:00
12:00
13:00
14:00
15:00
16:00
Tutorials
TUT18
Mammalian Physiology
PHOL0001-A4U-T1/2
Weeks: 6-9, 12
W3.04 - IOE - Bedford Way (20)
Lecture
Mammalian Physiology
PHOL0001-A4U-T1/2
Weeks: 6-10, 12-16, 20-23, 28
B40 Darwin LT - Darwin Building
Tutorials
PHOL0001-A4U-T1/2
Subgroup: 20 events
Lecture
Mammalian Physiology
PHOL0001-A4U-T1/2
Weeks: 6-10, 12-16, 20, 23-24, 27-28
Tutorials
PHOL0001-A4U-T1/2
Subgroup: 20 events

Tutorials

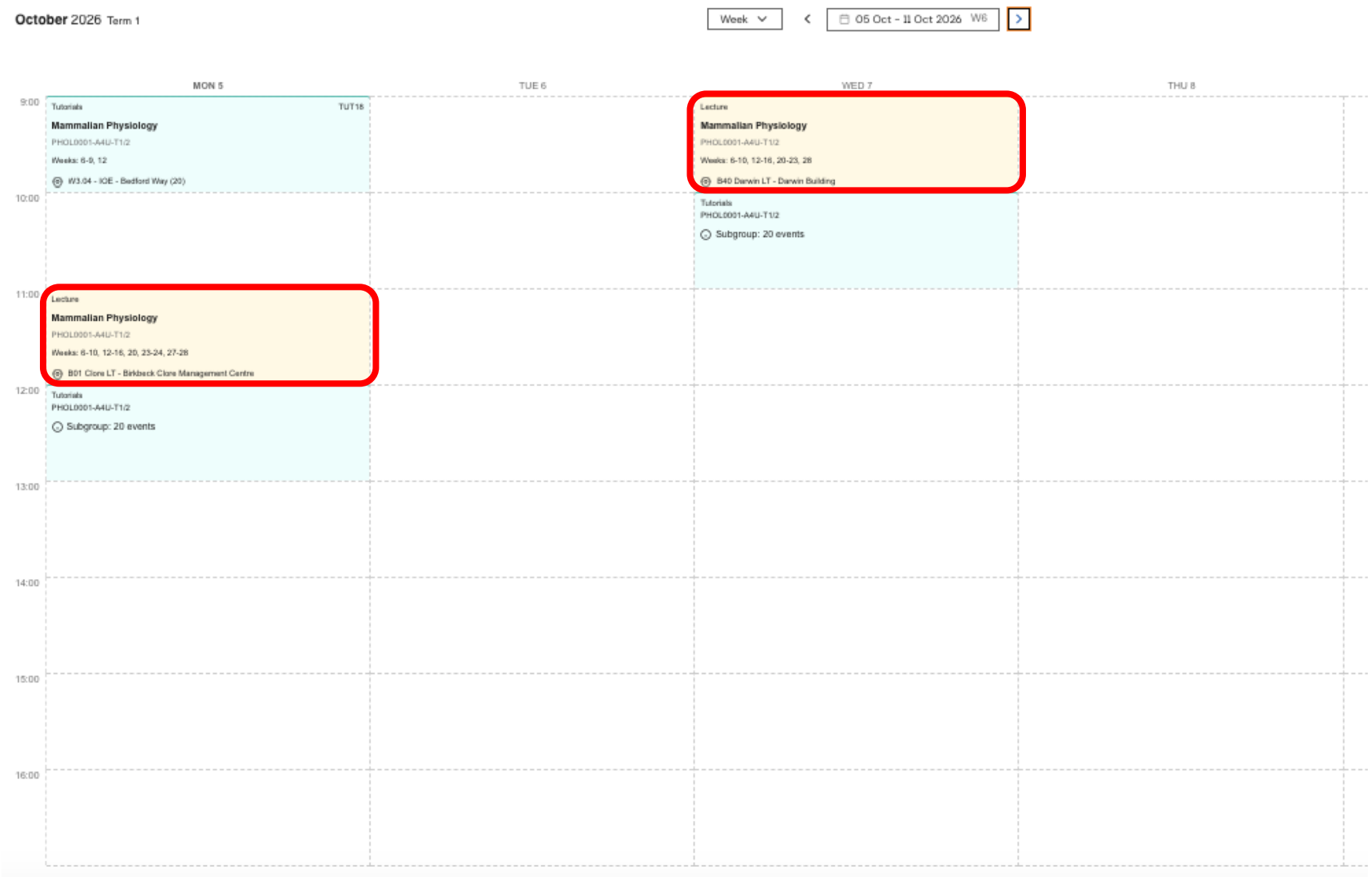
Teaching Events Timetable 2026-27

Term 1

Module	Day	Date	Time	Length	UCL Week	Synchronous Event Title	Lecturer	Venue
PHOL0001	Wed	30/09/2026	13:00	01:00	5	Tutorial A1&B1 Introduction	HOUSTON, Pam (Dr)	B01 Clore LT - Birkbeck Clore Management Centre
PHOL0001	Mon	05/10/2026	12:00	01:00	6	Tutorial A2 First group session	HOUSTON, Pam (Dr)	Various
PHOL0001	Wed	07/10/2026	10:00	01:00	6	Tutorial B2 First group session	HOUSTON, Pam (Dr)	Various
PHOL0001	Mon	19/10/2026	12:00	01:00	8	Tutorial A3 1-1 feedbback on abstract	HOUSTON, Pam (Dr)	Various
PHOL0001	Wed	21/10/2026	10:00	01:00	8	Tutorial B3 1-1 feedbback on abstract	HOUSTON, Pam (Dr)	Various
PHOL0001	Thu	05/11/2026	11:00	01:00	10	Tutorial A4 & B4 (inviglated essay)	PATEL, Jigisha (Dr)/ HOUSTON,PAM (Dr)	XLG2 Auditorium - Christopher Ingold Building
PHOL0001	Mon	09/11/2026	12:00	01:00	11	Tutorial A5 & 5B Belonging at UCL	HOUSTON, Pam (Dr)	TBC
PHOL0001	Mon	16/11/2026	12:00	01:00	12	Tutorial A61-1 Essay Feedback	HOUSTON, Pam (Dr)	Various
PHOL0001	Wed	18/11/2026	10:00	01:00	12	Tutorial B6 1-1 Essay Feedback	HOUSTON, Pam (Dr)	Various
PHOL0001	Mon	30/11/2026	12:00	01:00	14	Tutorial A7 & B7 Plagiarism &Academic Integrity	HOUSTON, Pam (Dr)	XLG2 Auditorium - Christopher Ingold Building
PHOL0001	Mon	14/12/2026	12:00	01:00	16	Tutorial A8 & B8 Key Skills	HOUSTON, Pam (Dr)	XLG2 Auditorium - Christopher Ingold Building

- Most of you will meet your personal academic tutor.
- If not, you should know when.
- Not heard, contact me (p.houston@ucl.ac.uk)

PHOL0001 timetable



PHOL0001 Practical classes (Thursday afternoons)

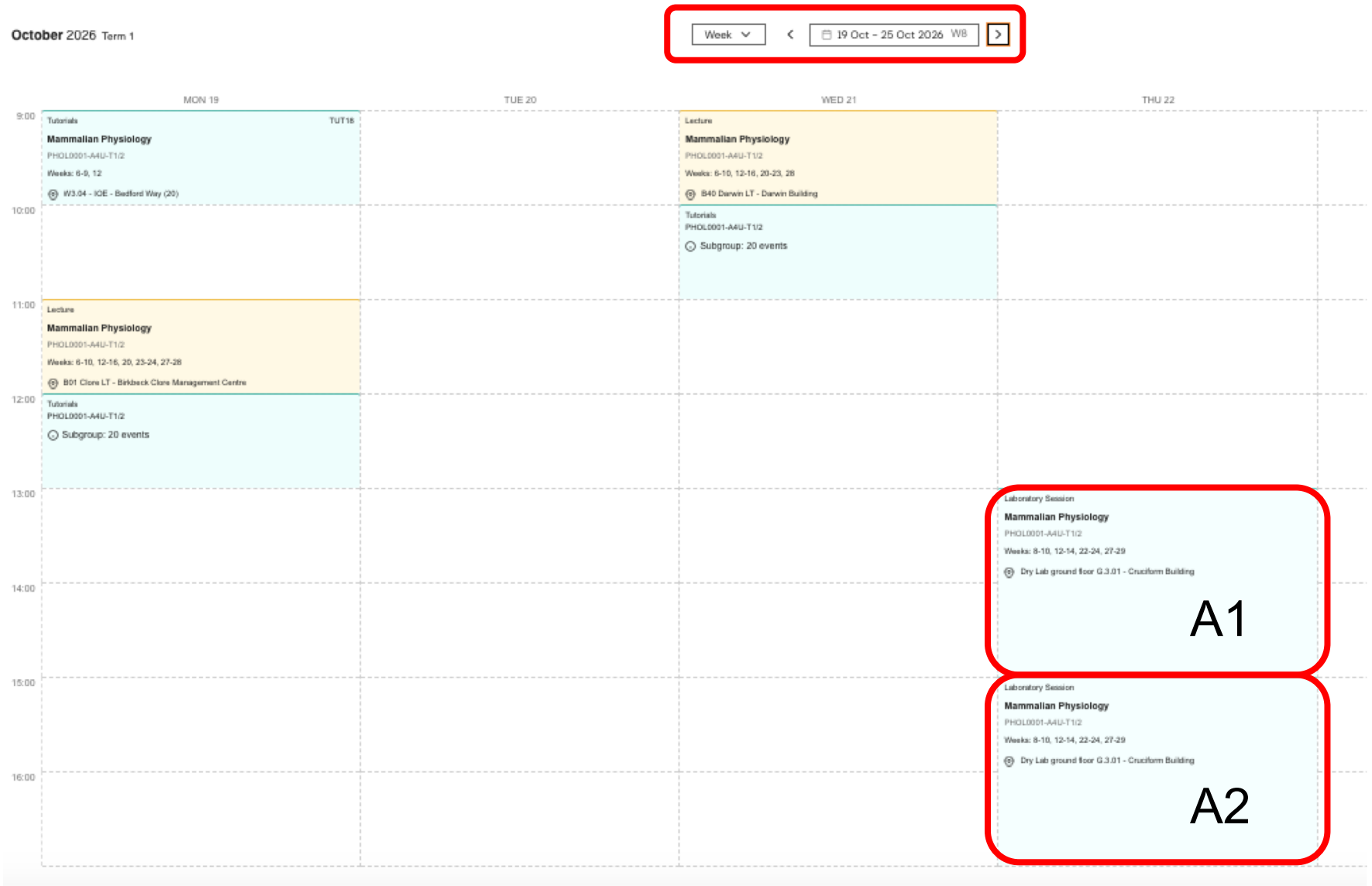
- Four laboratory based practical classes in the Cruciform building:
  - Term 1: Muscle and Breathing
  - Term 2: ECG and Exercise Physiology

- You are allocated one of six practical groups, A1, A2, B1, B2, C1 & C2. Groups are listed on Moodle.
- Each practical class is repeated six times over three consecutive weeks; two groups attend each afternoon.
- Make sure you take a note of the timing of your group's session i.e. A1 1-3pm, A2 3-5pm. All sessions show on the timetable, find your group.
- Turn up promptly at the time indicated and keep to the group you have been assigned to avoid overcrowding.

- Read the background information provided before attending your practical class and complete all pre-practical training and post-practical quizzes.
- You must complete all parts of a practical to be credited with the mark you achieve in the post-practical quiz on Moodle.
- You will need your ID card to get into the Cruciform building and to tap into the practical session
- You are not allowed to change session without prior permission from Dr Houston!
- You do not require any equipment, i.e. lab coats, laptop etc.
- Ensure you arrive 5 minutes before the session starts.

- I will run a “Getting started with Lt” session on Thursday with more information on practical classes.
- We use an online platform to deliver your practical classes called “LabTutor”, “Lt” for short.
- The training session is live online on Thursday 8th October 14:00pm, using Zoom.
- Try to login to Zoom before the session using SSO.
- This is not showing on your timetable, put it in your diary now. This for all groups.
- Before the session you will receive an invite from a company called AD Instruments, that is your Lt sign-up invite. There is no cost to you, UCL has registered you for this, but you need to accept. Look out for the email, this week.

Practical Timetable



B1/B2 week 9, C1/C2 week 10.

1. Tests 10% of module mark (4 x 2.5%)

Opens: Wednesday, 21 October 2026, 12:00 PM
Closes: Wednesday, 28 October 2026, 12:00 PM

Test 1 will be available here from 12:00 (Midday) on Wednesday 21st October 2026. You have to complete the assessment. Students who have an RAA will use the same link and the additional time will be programmed accordingly. This is the first of four on-line tests you are expected to attempt. **You have 3 attempts (2 hours each) and your highest score is recorded.** This test is worth 2.5% of the module overall.

- The test is a mixture of "single best answer", "calculated" and "matching" questions.
- Some questions are worth more marks than others. The mark value for each question is displayed.
- Do not spend a lot of time on a question that you do not know the answer to. Move on to the next question and go back to that question if you have time at the end. If you want to revisit a question afterwards, you can click the 'Flag question' option in the coloured box beside it.
- You will not be negatively marked for incorrect answers, so you should attempt to answer all questions.
- Some questions require a numerical answer. You will be told what units are expected but please do not write the units in your answer.
- You are permitted to use a UCL approved model calculator
- Moodle will save your answers each time you go to the next question.
- At the end of the test all attempts are automatically submitted.

Good luck!

Add question

Attempts allowed: 3

Grading method: Highest grade

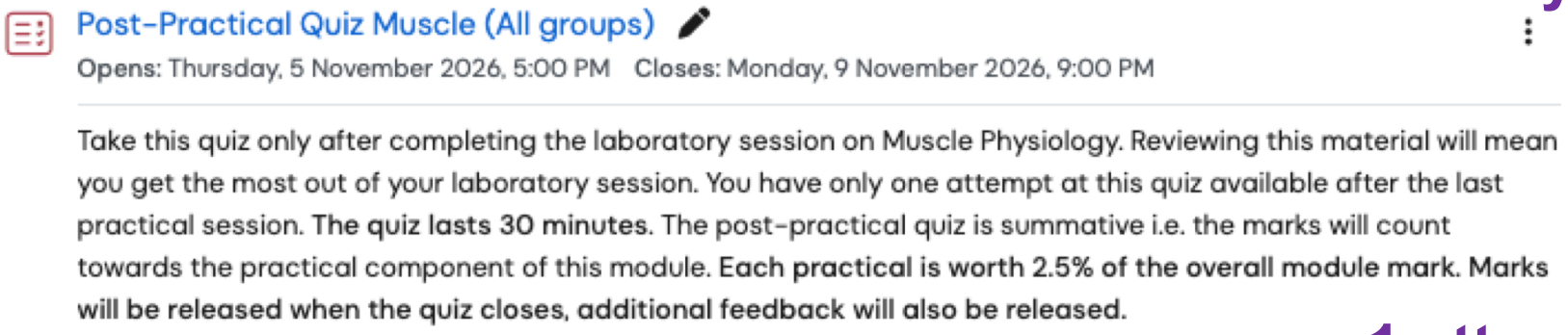
7 days

3 attempts

Rewards revision

2. Practical work 10% of module mark (4 x 2.5%)

4/5 days



1 attempt

Group A1/A2 week 8

Group B1/B2 week 9

Group C1/C2 week 10

knowledge application

- Post-practical muscle quiz opens
  - 5pm 05/11/26
- Post-practical muscle quiz closes
  - 9pm 09/11/26

1. Tests 10% of module mark (4 x 2.5%)

- Answer Single Best Answer (SBA) questions on pre-arranged topics
- Use quantitative data to make simple calculations

2. Practical work 10% of module mark (4 x 2.5%)

- Your practical marks will be based on how well you carry out your post-practical quizzes

3. Main exam 80% of module mark (1 x 80%)

- In the third term
- Online Moodle based, invigilated paper with three sections: SBA's, calculations and short answer questions.

Extenuating Circumstances and Mitagation

Extenuating Circumstances:

Where extenuating circumstances affects your academic work, you are able to request extensions or deferrals (and potentially other mitigation). Full details of this procedure can be found in the Academic Manual - this also includes details on how to apply, and what constitutes an extenuating circumstance. **Applications for mitigation are made online via Portico.**

Please note, module organisers cannot give extensions, you need to follow the above procedure and a departmental panel will make the decision.

Other support available for Life Sciences students

The Faculty Wellbeing Support team are here to support you throughout your degree.
Year 1 students can view this Moodle page for full details about the support offered from the Student Advisors.
Students in years 2, 3 and 4 can find details about the support offered by the Faculty Student Wellbeing team by following this link.

Students in other departments

If you are taking this module, but not from Life Sciences, the same EC procedure would apply to you, for additional support from your Faculty Wellbeing teams, please contact them directly.

Mitigation

Assessment Type code	Assessment Type name	Delayed Assessment Scheme: Mitigation
PD01 MD01	Practical examination (departmentally managed) or MCQ exam (departmentally managed)	"Other, see Moodle Page*" Multiple sub-components DAP can be used to exclude one (if not able to attend or write up.

Assessment Type code	Assessment Type name	Delayed Assessment Scheme: Mitigation
EC03	Exam (centrally managed, departmentally delivered)	Deferral without tuition to the next normal occasion (typically LSA period)

AI tools cannot be used in any of the PHOL0001 assessments

Use of AI tools in a PHOL0001 assessment will be considered Academic Misconduct

UCL AI Usage Categories:

On this page:

- Guidance
- Category 1: AI tools cannot be used
- Category 2: AI tools can be used in an assistive role
- Category 3: AI has an integral role
- UCL policy and guidance on acknowledging use of AI and referencing AI

Taken from https://www.ucl.ac.uk/teaching-learning/generative-ai-hub/using-ai-tools-assessment

Any Questions?

Claude Bernard and his pupils (1813–1878).



Source: http://wellcomeimages.org/indexplus/image/V0017769.html